

E-COMMERCE USER FUNNEL ANALYSIS REPORT

A Data Analytics Case Study

Program:

Data Analytics Mini Course Project by RevoU

Objective:

Analyzing user behavior data to identify engagement patterns and conversion opportunities.

Tools:

Python • Pandas • Matplotlib • Google Colab

Project Overview:

- User Engagement
- Acquisition Channel Performance
- Device Behavior
- Conversion Funnel Analysis

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1. Project Overview

This project analyzes e-commerce user interaction data to understand user behavior, engagement patterns, and conversion performance.

The analysis focuses on identifying user activity trends, understanding acquisition channel performance, evaluating device usage, and measuring the conversion journey from product search to add-to-cart action. The insights generated from this analysis can help businesses improve user experience, marketing strategies, and customer conversion opportunities.

2. Business Question

This project aims to answer the following questions:

- Which country has the highest number of active users during Q2 2025?
- How does monthly active user activity change across different acquisition channel?
- Which device is most frequently used based on user session?
- What percentage of users successfully move from search activity to adding items to their cart?

3. Dataset Description

The dataset contains user interaction events from an e-commerce platform.

The data includes:

- User information
- Session information
- Event timestamps
- Device types
- Acquisition channels
- User activity events

The main events analyzed in this project are:

- Search
- Add to Cart
- Other user interaction events

4. Methodology

The analysis was performed using Python with the Pandas library.

The main steps include:

Data Preparation

- Filtering relevant user activity data
- Removing duplicate records where necessary
- Transforming timestamps into usable date formats

Data Analysis

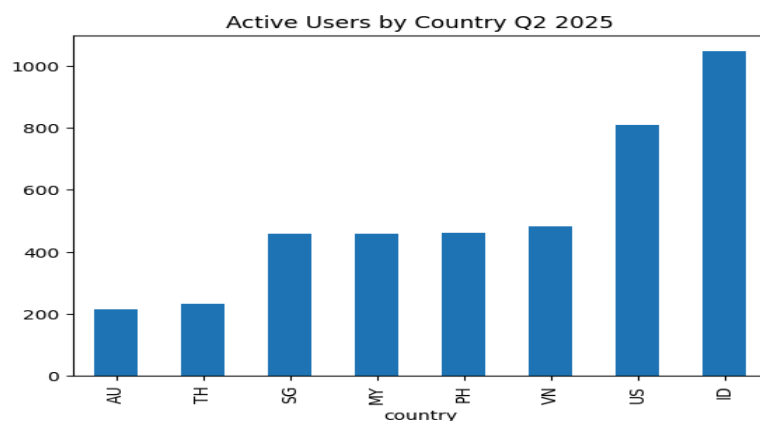
- Calculating unique active users
- Grouping users based on country and channel
- Counting unique sessions by device
- Measuring search-to-add-to-cart conversion rate

Data Visualization

Charts were created using Matplotlib to visualize user behavior patterns and analysis results.

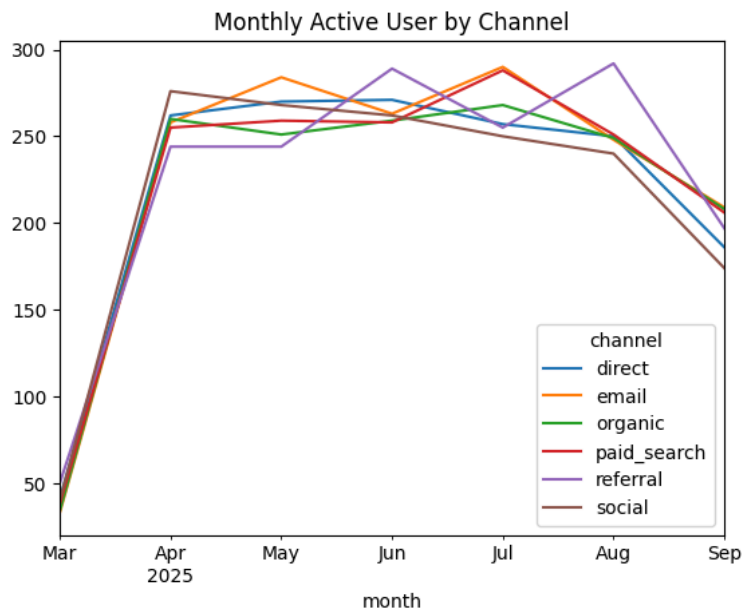
5. Analysis Results

5.1 Active Users by Country



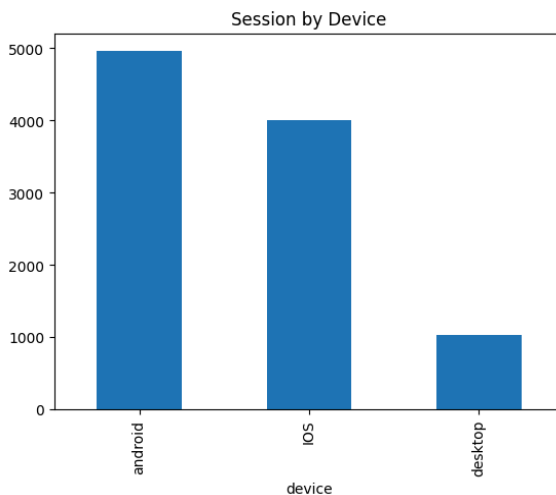
This analysis identifies the countries with the highest number of active users during Q2 2025. The country with the highest user activity is Indonesia. This indicates that this market contributes significantly to overall user engagement and may represent a valuable target for further business strategies.

5.2 Monthly Active Users by Channel



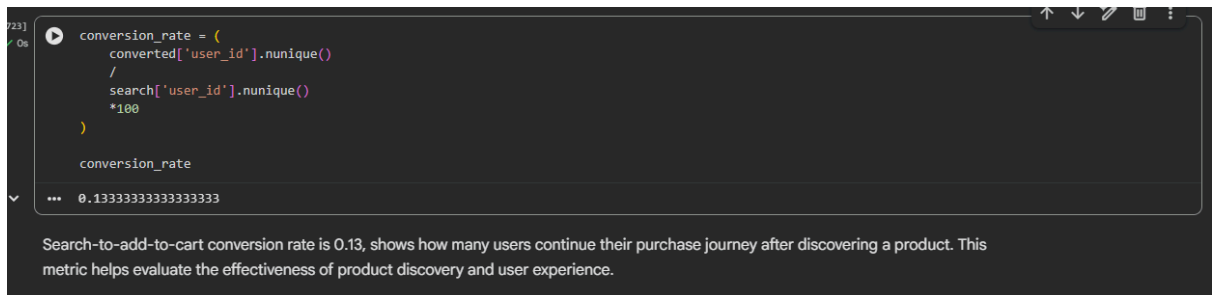
Monthly active users were analyzed across different acquisition channels to understand user engagement trends. The analysis shows differences in user activity patterns between channels. The channel with the strongest performance is Referral, suggesting that this channel effectively attracts and engages users.

5.3 Sessions by Device



Device usage was analyzed based on unique user sessions. The most frequently used device is Android. This suggests that users primarily interact with the platform through this device, making device optimization an important consideration for improving user experience.

5.4 Search-to-Add-to-Cart Conversion



```
223] In [ ]: conversion_rate = (
          converted['user_id'].nunique()
          /
          search['user_id'].nunique()
          *100
        )

        conversion_rate

Out[ ]: 0.13333333333333333
```

Search-to-add-to-cart conversion rate is 0.13, shows how many users continue their purchase journey after discovering a product. This metric helps evaluate the effectiveness of product discovery and user experience.

The conversion analysis measures the percentage of users who continue their shopping journey after searching for products.

CONVERSION RATE: 0.133 %

The conversion rate indicates how effectively product discovery activities lead users toward purchase intent. Improving search relevance, product information, and user experience may help increase conversion performance.

6. Recommendations

Based on the analysis, several recommendations can be considered:

- Improve High-Traffic User Experience
Focus on optimizing the experience for the most frequently used devices and channels.
- Optimize Marketing Channels
Further evaluate high-performing channels and allocate resources toward channels that generate stronger user engagement.
- Improve Conversion Journey
Analyze user drop-off points between product search and cart activity to increase successful conversions.
- Enhance Product Discovery
Improve search functionality and product presentation to encourage users to continue their purchase journey.

7. Conclusion

This analysis provides insights into e-commerce user behavior, engagement patterns, and conversion performance. By understanding where users come from, how they interact with the platform, and where conversion opportunities exist, businesses can make more data-driven decisions to improve customer experience and performance.

8. Tools & Project Link

Tools:

- Python
- Pandas
- Matplotlib
- Google Colab

Project Notebook:

<https://github.com/ayukrisnadi165-glitch/ecommerce-funnel-analysis.git>